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A small business invests $9,900 in equipment to produce a product. Each unit of the product costs $0.65 to produce and is sold for $1.20. How many units of the product must be sold before the revenue received equals the total expense of production, including the initial investment in equipment?

A. 12,000

B. 14,500

C. 15,230

D. 18,000

E. 20,000

Estimated Number of Home-Schooled Students by State, January 2001

State	Number (in thousands)
A	181
B	125
C	103
D	79
E	72

•

According to the table shown, the estimated number of home-schooled students in State A is approximately what percent greater than the number in State D?

A. 25%

B. 55%

C. 100%

D. 125%

E. 155%

•

When n liters of fuel were added to a tank that was already $\frac{1}{3}$ full, the tank was filled to $\frac{7}{9}$ of its capacity. In terms of n, what is the capacity of the tank, in liters?

A. $\frac{10}{9}n$

B. $\frac{4}{3}n$

C. $\frac{3}{2}n$

D. $\frac{9}{4}n$

E. $\frac{7}{3}n$

•

A bicycle race on a mountain trail began at an elevation 850 meters above Town X. The race course consisted of 5 phases, with the following elevation changes: a descent of 280 meters, an ascent of 350 meters, a descent of 620 meters, an ascent of 100 meters, and a descent of 400 meters. At what elevation, in meters above or below Town X, did the race end?

A. 850 below Town X

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A. 850 below Town X

B. Same elevation as Town X

C. 850 above Town X

D. 1,700 above Town X

E. 2,600 above Town X

•

The harvest yield from a certain apple orchard was 350 bushels of apples. If x of the trees in the orchard each yielded 10 bushels of apples, what fraction of the harvest yield was from these x trees?

A. $\frac{x}{35}$

B. $1 - \frac{x}{35}$

C. $10x$

D. $35 - x$

E. $350 - 10x$

•

In a certain fraction, the denominator is 16 greater than the numerator. If the fraction is equivalent to 80 percent, what is the denominator of the fraction?

A. 32

B. 64

C. 72

D. 80

E. 120

•

Jonathan, Kyle, and Michael agreed to pay equal shares of the monthly rent on an apartment. Because Jonathan has a cat, he paid a 1-time pet deposit of $250. The pet deposit and Jonathan's share of the monthly rent for the first month totaled $525. How much is the monthly rent on the apartment?

A. $1,575

B. $1,075

C. $1,025

D. $825

E. $775

•

The ratio of the amount of Alex's fuel oil bill for the month of February to the amount of his fuel oil bill for the month of January was $\frac{3}{2}$. If the fuel oil bill for February had been $40 more, the corresponding ratio would have been $\frac{5}{3}$ How much was Alex's fuel oil bill for January?

A. $240

B. $300

C. $360

D. $450

E. $540

•

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Al and Ben are drivers for SD Trucking Company. One snowy day, Ben left SD at 8:00 a.m. heading east and Al left SD at 11:00 a.m. heading west. At a particular time later that day, the dispatcher retrieved data from SD's vehicle tracking system. The data showed that, up to that time, Al had averaged 40 miles per hour and Ben had averaged 20 miles per hour. It also showed that Al and Ben had driven a combined total of 240 miles. At what time did the dispatcher retrieve data from the vehicle tracking system?

A. 1:00 p.m.

B. 2:00 p.m.

C. 3:00 p.m.

D. 5:00 p.m.

E. 6:00 p.m.

•

If $f = 3r\left(\frac{s}{1{,}000}\right)^2$ and if $s = 2{,}000$, what is r in terms of f?

A. $\frac{f}{6}$

B. $\frac{f}{12}$

C. $\frac{f}{36}$

D. $\frac{f^2}{36}$

E. $\frac{3f}{4}$

•

At the start of an experiment, a certain population consisted of 3 animals. At the end of each month after the start of the experiment, the population size was double its size at the beginning of that month. Which of the following represents the population size at the end of 10 months?

A. 2^3

B. 3^2

C. $2(3^{10})$

D. $3(2^{10})$

E. $3(10^2)$

•

The formula $F = \frac{9}{5C} + 32$ gives the relationship between the temperature in degrees Fahrenheit, F, and the temperature given in degrees Celsius, C. If the temperature is 85 degrees Fahrenheit, what is the temperature, to the nearest degree, in degrees Celsius?

A. 18°C

B. 23°C

C. 29°C

D. 47°C

E. 51°C

•

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•

If x and y are positive integers such that y is a multiple of 5 and $3x + 4y = 200$, then x must be a multiple of which of the following?

A. 3

B. 6

C. 7

D. 8

E. 10

•

Which of the following expressions can be written as an integer?

I. $\left(\sqrt{82} + \sqrt{82}\right)^2$

II. $(82)\left(\sqrt{82}\right)$

III. $\dfrac{\left(\sqrt{82}\right)\left(\sqrt{82}\right)}{82}$

A. None

B. I only

C. III only

D. I and II

E. I and III

•

Which of the following is equal to the average (arithmetic mean) of $(x + 2)^2$ and $(x - 2)^2$?

A. x^2

B. $x^2 + 2$

C. $x^2 + 4$

D. $x^2 + 2x$

E. $x^2 + 4x$

•

If $x^2 - 2 < 0$, which of the following specifies all the possible values of x?

A. $0 < x < 2$

B. $0 < x < \sqrt{2}$

C. $-\sqrt{2} < x < \sqrt{2}$

D. $-2 < x < 0$

E. $-2 < x < 2$

Book Number	Pages in Book	Total Pages Read
1	253	253
2	110	363
3	117	480
4	170	650
5	155	805

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6	50	855
7	205	1,060
8	70	1,130
9	165	1,295
10	105	1,400
11	143	1,543
12	207	1,750

•

Shawana made a schedule for reading books during 4 weeks (28 days) of her summer vacation. She has checked out 12 books from the library. The number of pages in each book and the order in which she plans to read the books are shown in the table above. She will read exactly 50 pages each day. The only exception will be that she will never begin the next book on the same day that she finishes the previous one, and therefore on some days she may read fewer than 50 pages. At the end of the 28th day, how many books will Shawana have finished?

A. 7

B. 8

C. 9

D. 10

E. 11

•

In Western Europe, x bicycles were sold in each of the years 1990 and 1993. The bicycle producers of Western Europe had a 42 percent share of this market in 1990 and a 33 percent share in 1993. Which of the following represents the decrease in the annual number of bicycles produced and sold in Western Europe from 1990 to 1993?

A. 9% of $\frac{x}{100}$

B. 14% of $\frac{x}{100}$

C. 75% of $\frac{x}{100}$

D. 9% of x

E. 14% of x

•

If k is a positive integer, what is the remainder when $(k+2)(k^3-k)$ is divided by 6?

A. 0

B. 1

C. 2

D. 3

E. 4

•

Which of the following fractions is closest to $\frac{1}{2}$?

A. $\frac{4}{7}$

B. $\frac{5}{9}$

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9

C. $\frac{6}{11}$

D. $\frac{7}{13}$

E. $\frac{9}{16}$

•

If $p \neq 0$ and $p - \frac{1-p^2}{p} = \frac{r}{p}$, then $r =$

A. $p + 1$

B. $2p - 1$

C. $p^2 + 1$

D. $2p^2 - 1$

E. $p^2 + p - 1$

•

If $\frac{|z|}{w} = 1$, which of the following must be true?

A. $z = -w$

B. $z = w$

C. $z^2 = w^2$

D. $z^2 = w^3$

E. $z^3 = w^3$

•

What number is 108 more than two-thirds of itself?

A. 72

B. 144

C. 162

D. 216

E. 324

•

A service provider charges c dollars for the first 50 hours of service used per month and 40 cents for each 30 minutes in excess of 50 hours used during the month. If x is an integer greater than 50, which of the following expressions gives this service provider's charge, in dollars, for a month in which x hours of service were used?

A. $c + 0.40x$

B. $c + 0.80x$

C. $c + 0.40(x - 50)$

D. $c + 0.80(x - 50)$

E. $c + 0.40(2x - 50)$

•

A salesperson who had been driving at a speed of 100 kilometers per hour slowed down to a speed of 47 kilometers

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A salesperson who had been driving at a speed of 100 kilometers per hour slowed down to a speed of 77 kilometers per hour. Approximately how many miles per hour was the speed reduced? (1 kilometer ≈ 0.625 mile)

A. 29

B. 33

C. 53

D. 63

E. 75

•

If $5x - 8 = 12$ and $x = y + 3$, then $y =$

A. $\frac{4}{5}$

B. 1

C. $\frac{17}{5}$

D. 4

E. 7

	Recording Time	Viewing Time
Tuesday	4 hours	None
Wednesday	None	1 to 2 hours
Thursday	2 hours	None
Friday	None	2 to 3 hours

•

The table above shows the numbers of hours of television programs that Jane recorded last week and the numbers of hours she spent viewing these recorded programs. No recorded program was viewed more than once. If h is the number of hours of recorded programs she had not yet viewed by the end of Friday, which of the following intervals represents all of the possible values of h?

A. $0 \leq h \leq 1$

B. $1 \leq h \leq 2$

C. $2 \leq h \leq 3$

D. $0 \leq h \leq 2$

E. $1 \leq h \leq 3$

•

A dance troupe has a total of 50 dancers split into 2 groups. The costumes worn by Group A cost $80 each, and those worn by Group B cost $90 each. If the total cost of all the costumes is $4,270, what is the total cost of the costumes worn by Group B?

A. $1,840

B. $2,070

C. $2,135

D. $2,160

E. $2,430

•

A doctor prescribed 18 cubic centimeters of a certain drug to a patient whose body weight was 120 pounds. If the typical dosage is 2 cubic centimeters per 15 pounds of body weight, by what percent was the prescribed dosage greater than the typical dosage?

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greater than the typical dosage?

A. 8%

B. 9%

C. 11%

D. 12.5%

E. 14.8%

•

The function f is defined by $f(x) = \sqrt{x} - 10$ for all positive numbers x. If $u = f(t)$ for some positive numbers t and u, what is t in terms of u?

A. $\sqrt{\sqrt{u} + 10}$

B. $(\sqrt{u} + 10)^2$

C. $\sqrt{u^2 + 10}$

D. $(u + 10)^2$

E. $(u^2 + 10)^2$

•

If 35% of a number is decreased by 15, then the result is 25% of the original number. What is the original number?

A. 1.5

B. 9

C. 25

D. 150

E. 900

•

A flower arrangement consists of 30 roses, each of which is either white or red. If a rose is to be selected at random from the flower arrangement, the probability that the rose selected will be white is twice the probability that it will be red. How many white roses are in the flower arrangement?

A. 5

B. 10

C. 15

D. 20

E. 25

•

Sixty percent of the people who responded to a company's advertisement for a job opening were interviewed over the phone. Twenty percent of those interviewed over the phone, that is, 3 people, were asked to come to the company for a personal interview. How many people responded to the advertisement?

A. 16

B. 20

C. 25

D. 40

E. 64

•

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$(-3)^{-2} =$

A. -9

B. -6

C. $-\frac{1}{9}$

D. $\frac{1}{9}$

E. 9

•

Mark bought a set of 6 flowerpots of different sizes at a total cost of $8.25. Each pot cost $0.25 more than the next one below it in size. What was the cost, in dollars, of the largest pot?

A. $1.75

B. $1.85

C. $2.00

D. $2.15

E. $2.30

•

If r and s are positive integers and $r - s = 6$, which of the following has the greatest value?

A. $2r$

B. $2s$

C. $r + s$

D. $2r - s$

E. $2s - r$

•

An auction house charges a commission of 15 percent on the first $50,000 of the sale price of an item, plus 10 percent on the amount of the sale price in excess of $50,000. What was the sale price of a painting for which the auction house charged a total commission of $24,000?

A. $115,000

B. $160,000

C. $215,000

D. $240,000

E. $365,000

•

At a certain college there are twice as many English majors as history majors and three times as many English majors as mathematics majors. What is the ratio of the number of history majors to the number of mathematics majors?

A. 6 to 1

B. 3 to 2

C. 2 to 3

D. 1 to 5

E. 1 to 6

•

A certain method for calculating pediatric drug dosages states that if d is the adult dosage, in milligrams, and y is the

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A certain method for calculating pediatric drug dosages states that if d is the adult dosage, in milligrams, and y is the child's age, in years, then the child's dosage, in milligrams, is $d\left(\frac{y+1}{24}\right)$. If the adult dosage for a particular drug is 800 milligrams, what is an 8-year-old child's dosage, in milligrams, according to this method?

A. 200

B. 225

C. 250

D. 275

E. 300

Questions 97 to 176 — Difficulty: Medium

97. The sides of each of two plastic cubes are numbered 1 through 6, and each number is equally likely to appear face up after either cube is rolled. What is the probability that after the two cubes are rolled, the sum of the two numbers appearing face up will be greater than 9?

A. $\frac{1}{6}$

B. $\frac{1}{5}$

C. $\frac{1}{4}$

D. $\frac{5}{18}$

E. $\frac{1}{3}$

98. On Saturday morning, Malachi will begin a camping vacation, and he will return home at the end of the first day on which it rains. If on the first three days of the vacation the probability of rain on each day is 0.2, what is the probability that Malachi will return home at the end of the day on the following Monday?

A. 0.008

B. 0.128

C. 0.488

D. 0.512

E. 0.640

99. According to a survey of 200 people, 60 enjoy skiing and 80 enjoy skating. If the number of people who enjoy neither skiing nor skating is 2 times the number of people who enjoy both skiing and skating, how many people surveyed enjoy neither skiing nor skating?

A. 20

B. 40

C. 50

D. 80

E. 120

$$m \oplus p = n$$
$$n \oplus r = m$$
$$n \oplus q = q$$
$$p \oplus q = p$$
$$q \oplus p = r$$

100. If the relations shown hold for the operation $\oplus$ and the numbers m, n, p, q, and r, then $[(m \oplus n) \oplus q] \oplus p =$

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100. If the relations shown hold for the operation $\oplus$ and the numbers m, n, p, q, and r, then $[(m \oplus p) \oplus q] \oplus p =$

A. m

B. n

C. p

D. q

E. r

101. To rent a tractor, it costs a total of x dollars for the first 24 hours, plus y dollars per hour for each hour in excess of 24 hours. Which of the following represents the cost, in dollars, to rent a tractor for 36 hours?

A. $x + 12y$

B. $x + 36y$

C. $12x + y$

D. $24x + 12y$

E. $24x + 36y$

102. If the mass of 1 cubic centimeter of a certain substance is 7.3 grams, what is the mass, in kilograms, of 1 cubic meter of this substance? (1 cubic meter = 1,000,000 cubic centimeters; 1 kilogram = 1,000 grams)

A. 0.0073

B. 0.73

C. 7.3

D. 7,300

E. 7,300,000

103. If $z \neq 0$ and $z + \dfrac{1 - 2z^2}{z} = \dfrac{w}{z}$, then $w =$

A. $z + 1$

B. $z^2 + 1$

C. $-z^2 + 1$

D. $-z^2 + z + 1$

E. $-2z^2 + 1$

104. For all real numbers a, b, c, d, e, and f, the operation Θ is defined by the equation $(a, b, c)\Theta(d, e, f) = ad + be + cf$. What is the value of $(1, -2, 3)\Theta(1, \frac{1}{2}, \frac{1}{3})$?

A. −1

B. $\frac{5}{6}$

C. 1

D. $\frac{5}{2}$

E. 3

ABSENCES FROM A CERTAIN MATH CLASS DURING THE SPRING SEMESTER	
Number of Absences	Number of Students
0	4
1	3
2	10

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2	10
3	3
4	5
5 or more	3

105. The table above shows the distribution of the number of absences from a certain Math class during the spring semester. For those students who had at least 1 absence, what was the median number of absences?

A. 1.5

B. 2

C. 2.5

D. 3

E. 3.5

106. If $\frac{x}{y} = \frac{c}{d}$ and $\frac{d}{c} = \frac{b}{a}$, which of the following must be true?

I. $\frac{y}{x} = \frac{b}{a}$

II. $\frac{x}{a} = \frac{y}{b}$

III. $\frac{y}{a} = \frac{x}{b}$

A. I only

B. II only

C. I and II only

D. I and III only

E. I, II, and III

107. If $[x]$ denotes the least integer greater than or equal to x and $\left[\frac{x}{2}\right] = 0$, which of the following could be the value of x?

A. –2

B. $-\frac{3}{2}$

C. $\frac{1}{2}$

D. 1

E. 2

108. If $a(a+2) = 24$ and $b(b+2) = 24$, where $a \neq b$, then $a + b =$

A. –48

B. –2

C. 2

D. 46

E. 48

109. In a recent election, Ms. Robbins received 8,000 votes cast by independent voters, that is, voters not registered with a specific political party. She also received 10 percent of the votes cast by those voters registered with a political party. If N is the total number of votes cast in the election and 40 percent of the votes cast were cast by independent voters, which of the following represents the number of votes that Ms. Robbins received?

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independent voters, which of the following represents the number of votes that Ms. Robbins received?

A. $0.06N + 3,200$

B. $0.1N + 7,200$

C. $0.4N + 7,200$

D. $0.1N + 8,000$

E. $0.06N + 8,000$

110. The profit P, in dollars, for any given month at a certain company is defined by $P = I - C$, where I represents total income, in dollars, and C represents total costs, in dollars, for the month. For each of the first 4 months of the year, $C = I + 32,000$; and for each of the next 3 months, $I = C + 36,000$. If $I = C + 10,000$ for each of the 5 remaining months of the year, what was the company's total profit for the 12-month year?

A. $10,000

B. $30,000

C. $40,000

D. $50,000

E. $70,000

111. A manufacturer makes and sells 2 products, P and Q. The revenue from the sale of each unit of P is $20.00 and the revenue from the sale of each unit of Q is $17.00. Last year the manufacturer sold twice as many units of Q as P. What was the manufacturer's average (arithmetic mean) revenue per unit sold of these 2 products last year?

A. $28.50

B. $27.00

C. $19.00

D. $18.50

E. $18.00

112. A worker carries jugs of liquid soap from a production line to a packing area, carrying 4 jugs per trip. If the jugs are packed into cartons that hold 7 jugs each, how many jugs are needed to fill the last partially filled carton after the worker has made 17 trips?

A. 1

B. 2

C. 4

D. 5

E. 6

113. Last year a state senate consisting of only Republican and Democrat members had 20 more Republican members than Democrat members. This year the senate has the same number of members as last year, but it has 2 fewer Republican members than last year.

If this year the number of Republican members is $\frac{2}{3}$ the number of senate members, how many members does the senate have this year?

A. 33

B. 36

C. 42

D. 45

E. 48

114. Sam has $800 in his account. He will deposit $1 in his account one week from now, $2 two weeks from now,

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114. Sam has $800 in his account. He will deposit $1 in his account one week from now, $2 two weeks from now, and each week thereafter he will deposit an amount that is $1 greater than the amount that he deposited one week before. If there are no other transactions, how much money will Sam have in his account 50 weeks from now?

A. $850

B. $1,200

C. $1,675

D. $2,075

E. $3,350

115. A certain state's milk production was 980 million pounds in 2007 and 2.7 billion pounds in 2014. Approximately how many more million gallons of milk did the state produce in 2014 than in 2007? (1 billion = 10^9 and 1 gallon = 8.6 pounds.)

A. 100

B. 200

C. 1,700

D. 8,200

E. 14,800

116. Working simultaneously and independently at an identical constant rate, four machines of a certain type can produce a total of x units of Product P in 6 days. How many of these machines, working simultaneously and independently at this constant rate, can produce a total of $3x$ units of Product P in 4 days?

A. 24

B. 18

C. 16

D. 12

E. 8

117. The symbol Δ denotes one of the four arithmetic operations: addition, subtraction, multiplication, or division. If 6 Δ 3 ≤ 3, which of the following must be true?

I. 2 Δ 2 = 0

II. 2 Δ 2 = 1

III. 4 Δ 2 = 2

A. I only

B. II only

C. III only

D. I and II only

E. I, II, and III

118. If $mn \neq 0$ and 25 percent of n equals $37\frac{1}{2}$ percent of m, what is the value of $\frac{12n}{m}$?

A. 18

B. $\frac{32}{3}$

C. 8

D. 3

E. $\frac{9}{8}$

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119. N and M are each 3-digit integers. Each of the numbers 1, 2, 3, 6, 7, and 8 is a digit of either N or M. What is the smallest possible positive difference between N and M ?

A. 29

B. 49

C. 58

D. 113

E. 131

Technique	Percent of Consumers
Television ads	35%
Coupons	22%
Store displays	18%
Samples	15%

120. The table shows partial results of a survey in which consumers were asked to indicate which one of six promotional techniques most influenced their decision to buy a new food product. Of those consumers who indicated one of the four techniques listed, what fraction indicated either coupons or store displays?

A. $\frac{2}{7}$

B. $\frac{1}{3}$

C. $\frac{2}{5}$

D. $\frac{4}{9}$

E. $\frac{1}{2}$

121.

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